第三章 相变的热力学理论

§1

1. 系统只有一种粒子：$dG = -SdT + VdP$

G 为广延量

化学势

对于开系：$dG = -SdT + VdP + \mu dn$

$$G(T,P,n) = n \cdot g(T,P)$$

$$\rightarrow \mu = \left(\frac{\partial G}{\partial n}\right)_{T,P} = g(T,P)$$

$$S = -\left(\frac{\partial G}{\partial T}\right)_{P,n}, \quad V = \left(\frac{\partial G}{\partial P}\right)_{T,n}, \quad \mu = \left(\frac{\partial G}{\partial n}\right)_{T,P}$$

$U = G + TS - PV \qquad dU = TdS - PdV + \mu dn$

$H = G + TS \qquad dH = TdS + VdP + \mu dn$

$F = G - PV \qquad dF = -SdT - PdV + \mu dn$

2. 巨热力势

$$J = F - \mu n$$

$$dJ = -SdT - PdV - nd\mu$$

3. 热动平衡的条件

(1) 熵判据

$\delta S = 0, \quad \delta^2 S < 0, \quad \delta U = \delta V = \delta N = 0$

两个系统：

$S = S_1 + S_2, \quad U = U_1 + U_2, \quad N = N_1 + N_2, \quad V = V_1 + V_2$

$\rightarrow \delta U_1 = -\delta U_2, \quad \delta N_1 = -\delta N_2, \quad \delta V_1 = -\delta V_2$

$$dS = \frac{1}{T}(dU + PdV - \mu dn)$$

$$\delta S = \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\delta U_1 + \left(\frac{P_1}{T_1} - \frac{P_2}{T_2}\right)\delta V_1 - \left(\frac{\mu_1}{T_1} - \frac{\mu_2}{T_2}\right)\delta N_1 = 0$$

$\Rightarrow T_1 = T_2, \quad P_1 = P_2, \quad \mu_1 = \mu_2$

$\delta U = \delta V = 0$

$dS = -\frac{1}{T}(\mu_1 - \mu_2)\delta N_1 > 0 \rightarrow \mu_1 > \mu_2, \quad \delta N_1 < 0$

化学势高 → 物质减少.

(2) 粒子数不守恒.

$\delta S = 0 \to \mu_1 = \mu_2 = 0$ $\Rightarrow$ 化学势 $= 0$

4. 平衡的稳定条件.

$\delta U = 0$, $\delta^2 U > 0$, $\delta S = \delta V = \delta N = 0$

$\delta^2 S_1 = -\delta^2 S_2 \cdots$

$$\delta U = \sum_\alpha T_\alpha \delta S_\alpha - P_\alpha \delta V_\alpha + \mu_\alpha \delta N_\alpha$$

$$\delta^2 U = \underbrace{\sum_\alpha (T_\alpha \delta^2 S_\alpha - P_\alpha \delta^2 V_\alpha + \mu_\alpha \delta^2 N_\alpha)}_{=0} + \delta T_\alpha \delta S_\alpha - \delta P_\alpha \delta V_\alpha + \delta \mu_\alpha \delta N_\alpha$$

$$= \sum_\alpha \delta T_\alpha \delta S_\alpha - \delta P_\alpha \delta V_\alpha + \delta \mu_\alpha \delta N_\alpha$$

记 $S_\alpha = N_\alpha s_\alpha$, $V_\alpha = N_\alpha v_\alpha$

$$= \sum_\alpha (s_\alpha \delta T_\alpha - v_\alpha \delta P_\alpha + \delta \mu_\alpha) \delta N_\alpha + (\delta T_\alpha \delta s_\alpha - \delta P_\alpha \delta v_\alpha) N_\alpha$$

$$= \sum_\alpha (\delta T_\alpha \delta s_\alpha - \delta P_\alpha \delta v_\alpha) N_\alpha > 0$$

$s = s(T, v)$, $P = P(T, v)$ $\quad d(U - TS)$

$\delta s = \left(\frac{\partial s}{\partial T}\right)_v \delta T + \left(\frac{\partial s}{\partial v}\right)_T \delta v$ $\quad = -SdT - PdV + \mu dN$

$\delta P = \left(\frac{\partial P}{\partial T}\right)_v \delta T + \left(\frac{\partial P}{\partial v}\right)_T \delta v$ $\quad \to \left(\frac{\partial s}{\partial v}\right)_T = \left(\frac{\partial P}{\partial T}\right)_v$

$$\hookrightarrow \delta^2 U = \sum_\alpha \left[\delta T_\alpha \left(\frac{\partial s}{\partial T}\right)_v \delta T_\alpha + \delta T_\alpha \left(\frac{\partial s}{\partial v}\right)_T \delta v_\alpha - \delta v_\alpha \left(\frac{\partial P}{\partial T}\right)_v \delta T_\alpha - \delta v_\alpha \left(\frac{\partial P}{\partial v}\right)_T \delta v_\alpha\right]$$

$$= \sum_\alpha \left(\frac{\partial s}{\partial T}\right)_v (\delta T_\alpha)^2 - \left(\frac{\partial P}{\partial v}\right)_T (\delta v_\alpha)^2$$

$$= \sum_\alpha \frac{C_{v\alpha}}{T} (\delta T_\alpha)^2 - \left(\frac{\partial P}{\partial v}\right)_T (\delta v_\alpha)^2 > 0$$

$\Rightarrow C_v > 0$, $\left(\frac{\partial P}{\partial v}\right)_T < 0$ $\quad \kappa_T = -\frac{1}{v}\left(\frac{\partial v}{\partial P}\right)_T > 0$ 等温压缩系数.

类似地: $T = T(s, P)$, $V = V(s, P)$

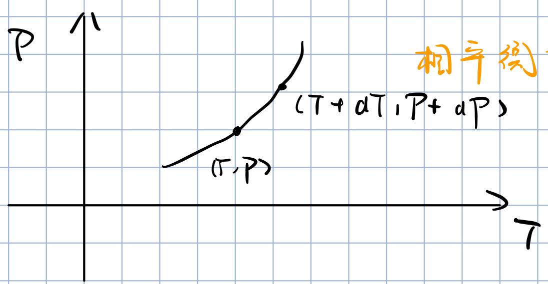
⋮

$\Rightarrow C_p > 0$, $\left(\frac{\partial v}{\partial P}\right)_s < 0$, $\kappa_s = -\frac{1}{v}\left(\frac{\partial v}{\partial P}\right)_s > 0$

§7 单元系的复相平衡

α, β 两相系，相平衡

$T^\alpha = T^\beta = T,\quad p^\alpha = p^\beta = p,\quad \mu^\alpha(T,p) = \mu^\beta(T,p) = \mu(T,p)$



$$d\mu^\alpha = d\mu^\beta = d\mu = -S\,dT + V\,dP$$

$$-S^\alpha dT + V^\alpha dP = -S^\beta dT + V^\beta dP$$

$$\to \frac{dP}{dT} = \frac{S^\beta - S^\alpha}{V^\beta - V^\alpha}$$

相变潜热：$L = \lambda_{\alpha\beta} = T(S_\beta - S_\alpha)$

蒸气压：α - 凝聚相，β - 气相。

$$PV^\beta = RT,\quad V^\alpha \ll V^\beta$$

$$\frac{dP}{dT} = \frac{L}{T(V^\beta - V^\alpha)} \approx \frac{L}{TV^\beta} \approx \frac{PL}{RT^2}$$

$$\to \frac{1}{P}\frac{dP}{dT} = \frac{L}{RT^2} \qquad \ln P - \ln P_0 = -\frac{L}{R}(T - T_0)$$

解得 $P = P_0 e^{-\frac{L}{R}\left(\frac{1}{T} - \frac{1}{T_0}\right)}$

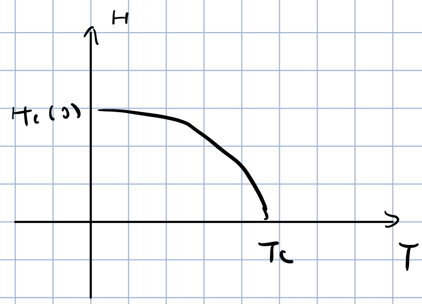
§6 正常-超导相变的热力学理论

1. 超导

$\vec{B} = \mu_0(\vec{H} + \vec{m}) = 0$

临界磁场 H_c：$H > H_c$ 时，超导被破坏。

$$H_c(T) = H_c(0) \cdot \left[1 - \left(\frac{T}{T_c}\right)^2\right]$$



2. G 与 H 关系

$dU = TdS - pdV + VHdB$

$V = 1$，$N \leftrightarrow S$ 相变中 $dV \ll 1$

$dU = TdS + HdB$

$G = U - TS - HB$

$dG = -SdT - BdH$，$\left(\frac{\partial G}{\partial H}\right)_T = -B$

$G(T,H) - G(T,0) = -\int_0^H B\,dH$

金属：$\vec{m} = 0$，$\vec{B} = \mu_0 \vec{H}$

$\rightarrow$ $G_n(T,H) = G_n(T,0) - \frac{1}{2}\mu_0 \vec{H}^2$

$G_s(T,H) = G_s(T,0)$

$\hookrightarrow$ $G_s(T,H) - G_n(T,H) = G_s(T,0) - G_N(T,0) + \frac{1}{2}\mu_0 \vec{H}^2$

N-S 相平衡：$\mu_N(T,H_c) = \mu_s(T,H_c) \Leftrightarrow G_N(T,H_c) = G_s(T,H_c)$

$\Rightarrow$ $G_s(T,H) - G_N(T,H) = \frac{\mu_0}{2}\left[H^2 - H_c^2(T)\right]$

$$\begin{cases} H > H_c：G_N < G_s，N \text{ 相稳定.} \\ H < H_c：G_s > G_N，S \text{ 相稳定.} \end{cases}$$

3. 平衡曲线的克拉珀龙方程.

$dG_N = -S_N\,dT - B_N\,dH$, $dG_S = -S_S\,dT - B_S\,dH$

曲线上：$dG_N = dG_S$

$\hookrightarrow \frac{dH_c}{dT} = -\frac{S_S - S_N}{B_S - B_N} = \frac{1}{\mu_0 H_c}(S_S - S_N)$ 　 $B_N = \mu_0 H\ (H_c)$, $B_S = 0$

相变潜热：$L_{SN} = T(S_S - S_N) = \mu_0 H_c T \frac{dH_c}{dT} < 0$

$\Rightarrow$ $S_S < S_N$: 超导态更有序.

$L_{SN} < 0$: $N \to S$ 相放热.

1级相变：$H_c \neq 0$, $L_{SN} \neq 0$

2级相变：$H_c = 0$, $L_{SN} = 0$

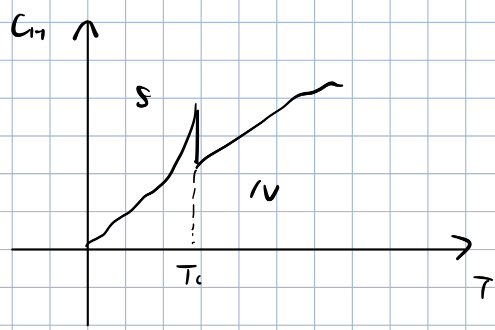
4. $C_H = T\left(\frac{\partial S}{\partial T}\right)_H$

$C_{HS} - C_{HN} = T\cdot\left(\frac{\partial}{\partial T}\right)_H\left(\mu_0 H_c \frac{dH_c}{dT}\right)$

$= \mu_0 T\left[\left(\frac{dH_c}{dT}\right)^2 + H_c \frac{d^2 H_c}{dT^2}\right]$

$T = T_c$ 时，$H_c = 0$;

$\Delta H = \mu_0 T\left(\frac{dH_c}{dT}\right)^2 > 0$



§7 相变的分类 Ehrenfest 方程.

1. 一级相变.

存在潜热, $L = T(S^{(2)} - S^{(1)})$ 和比容突变. $\Delta V = (V^{(2)} - V^{(1)})$

$\mu^{(1)}(T,P) = \mu^{(2)}(T,P)$, $\frac{\partial \mu^{(1)}}{\partial T} \neq \frac{\partial \mu^{(2)}}{\partial T}$, $\frac{\partial \mu^{(1)}}{\partial P} \neq \frac{\partial \mu^{(2)}}{\partial P}$

化学势 μ 的一阶偏导数存在突变.

2. n 级相变

化学势 μ 的 n 阶偏导数存在突变.

e.g. 2级相变

$S = -\frac{\partial \mu}{\partial T}$, $V = \frac{\partial \mu}{\partial P}$ 不存在突变,

$C_P = T\left(\frac{\partial S}{\partial T}\right)_P = -T\frac{\partial^2 \mu}{\partial T^2}$, $\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P = \frac{1}{V}\frac{\partial^2 \mu}{\partial T \partial P}$,

$\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T = -\frac{1}{V}\frac{\partial^2 \mu}{\partial P^2}$

3. Ehrenfest 方程.

一级相变时: $\frac{dP}{dT} = \frac{\Delta S}{\Delta V}$ $\qquad$ $dG_1 = -S_1 dT + V_1 dP = dG_2 = -S_2 dT + V_2 dP$

$\Delta S, \Delta V \to 0$, $\qquad$ $\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$

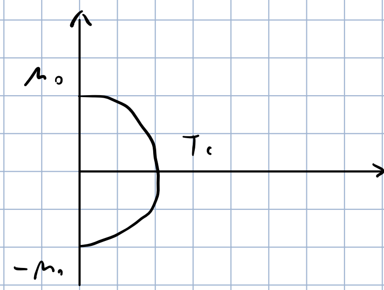
① $$\frac{dP}{dT} = \frac{\left(\frac{\partial S^\beta}{\partial T}\right)_P - \left(\frac{\partial S^\alpha}{\partial T}\right)_P}{\left(\frac{\partial V^\beta}{\partial T}\right)_P - \left(\frac{\partial V^\alpha}{\partial T}\right)_P} = \frac{C^\beta - C^\alpha}{TV(\alpha^\beta - \alpha^\alpha)} = \frac{\Delta C_P}{TV\Delta\alpha}$$

② $$\frac{dP}{dT} = \frac{\left(\frac{\partial S^\beta}{\partial P}\right)_T - \left(\frac{\partial S^\alpha}{\partial P}\right)_T}{\left(\frac{\partial V^\beta}{\partial P}\right)_T - \left(\frac{\partial V^\alpha}{\partial P}\right)_T} = \frac{\Delta\alpha}{\Delta\kappa_T}$$ $\qquad$ $\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$, $\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T$

$\Rightarrow \Delta C_P = TV\frac{(\Delta\alpha)^2}{\Delta\kappa_T}$

§ 8 Landau 二级相变理论.

序参量，对称性破缺



单轴铁磁体：序参量 - 磁化强度 $m = m(T)$

Landau 自由能：

$$F(T, m) = \sum_{n=0}^{\infty} a_n m^n$$

$H=0$ 时，对称性要求 $F(T,m) = F(T,-m)$

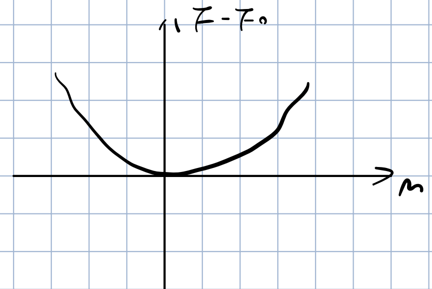
$$\hookrightarrow F(T,m) = F_0(T) + a_2(T) m^2 + a_4(T) m^4$$

$$\frac{\partial F}{\partial m} = m(2a_2 + 4a_4 m^2) = 0$$

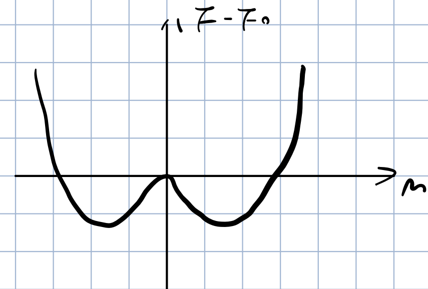
$$\frac{\partial^2 F}{\partial m^2} = 2a_2 + 12 a_4 m^2 > 0$$

① $m = 0$，无序态.，$T > T_c$，$a_2 > 0$

② $m = \pm\sqrt{-\frac{a_2}{2a_4}}$，有序态，$T < T_c$，$a_2 < 0$



① $T > T_c$



① $T < T_c$

$T = T_c$ 时：$a_2 = 0$

T_c 附近：设 $a_2 = a_{20}(T - T_c) = a_{20}t$，$a_{20} > 0$

$$a_4 = a_{40}$$

$$\hookrightarrow m = \begin{cases} 0, & t > 0 \\ \pm\sqrt{\frac{a_{20}}{2a_{40}}}(-t)^{1/2}, & t < 0 \end{cases}$$

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3. 零磁场的熵.

展开

$$S = -\left(\frac{\partial F}{\partial T}\right)_m = -\frac{\partial F_0}{\partial T} - a_{20} m^2 = -S_0(T) - a_{20} m^2$$

$$\Rightarrow S = \begin{cases} S_0(T), & T > T_c \\ S_0(T) - \frac{a_{20}}{2a_4}(T_c - T), & T < T_c \end{cases}$$

4. 外磁场不为零时的比热突变.

$G = F + PV$，磁介质系统：$V \Rightarrow m$，$P \to -\mu_0 H$、

$$G(T, H, m) = F(T, m) - \mu_0 H m$$

$$= F_0(T) + a_2(T) m^2 + a_4(T) m^4 - \mu_0 H m$$

$$\left(\frac{\partial G}{\partial m}\right)_{T,H} = \left(\frac{\partial F}{\partial m}\right)_{T,H} - \mu_0 H = 2a_2(T) m + 4a_4(T) m^3 - \mu_0 H = 0,$$

临界：$a_2 = a_{20}(T - T_c)$，$a_4 = a_4$

$\Rightarrow T = T_c$ 时，$a_2 = 0$，$m = \left(\frac{\mu_0 H}{4a_4}\right)^{1/3}$

$$\left(\frac{\partial^2 G}{\partial m^2}\right)_{T,H} = 2a_2(T) + 12 a_4 m^2 \geq 0$$

定义磁化率：$\chi = \left(\frac{\partial m}{\partial H}\right)_T$.

对 $\left(\frac{\partial G}{\partial m}\right)_{T,H} = 0$ 求导：$\chi(2a_2 + 12a_4 m^2) = \mu_0$

$$\chi = \frac{\mu_0}{2a_2 + 12a_4 m^2}$$

在临界：$\chi^0 = \chi|_{H \to 0}$

$$= \begin{cases} \frac{\mu_0}{2a_2(T)} = \frac{\mu_0}{2a_{20}} \frac{1}{T - T_c}, & T > T_c \\ -\frac{\mu_0}{4a_2(T)} = \frac{\mu_0}{4a_{20}} \frac{1}{T_c - T}, & T < T_c \end{cases}$$

$$C_H - C_m = -\mu_0 T \cdot \left(\frac{\partial H}{\partial T}\right)_m \left(\frac{\partial m}{\partial T}\right)_H \qquad H = \frac{1}{\mu_0}\left[2a_2(T) m + 4a_4 m^3\right]$$

$$= -\mu_0 T \cdot \frac{2a_{20} m}{\mu_0} \cdot \left(\frac{\partial m}{\partial T}\right)_H$$

$$= -2a_{20} m T \left(\frac{\partial m}{\partial T}\right)_H$$

$H \to 0$ 极限：

$$C_H^0 - C_m^0 = -2a_{20}T \left(m \left(\frac{\partial m}{\partial T}\right)_H\right)_{H\to 0}$$

$$2a_{20}(T-T_c)\,m + 4a_4 m^3 = 0 \quad (H=0)$$

$$m = \begin{cases} 0, & T > T_c \\ \sqrt{\dfrac{a_{20}}{2a_4}}\,(T_c - T)^{1/2}, & T < T_c \end{cases}$$

$$\frac{\partial m}{\partial T} = \begin{cases} 0 \\ -\dfrac{1}{2}\sqrt{\dfrac{a_{20}}{2a_4}}\,(T_c - T)^{-1/2} = -\dfrac{a_{20}}{4a_4}\dfrac{1}{m} \end{cases}$$

$$C_H^0 = \begin{cases} C_m^0, & T > T_c \\ C_m^0 + \dfrac{a_{20}^2 T}{2a_4}, & T < T_c \end{cases}$$

在临界点附近，C_H^0 有一跃变：

$$\Delta C_H^0 = \frac{a_{20}^2 T}{2a_4}$$

§9 临界现象.

临界指数：在临界点附近，热力学函数与关联函数的行为.

$$f(\varepsilon) = A\varepsilon^{\lambda}\{1 + B\varepsilon^{x} + \cdots\} \quad (x>0)$$

热力学量， e.g. $\varepsilon = \frac{T-T_c}{T_c}$

当 $\varepsilon \to 0$ 时， $f(\varepsilon) \to A\varepsilon^{\lambda} \sim \varepsilon^{\lambda}$

定义 $\lambda = \lim_{\varepsilon\to 0} \frac{\ln f(\varepsilon)}{\ln \varepsilon}$ ，为临界指数.

1. 临界指数 β

$$m(T) \sim (T_c - T)^{\beta} \; ; \; (T \to T_c^-, H = 0)$$

$$(\rho_l - \rho_g) \sim (T_c - T)^{\beta} \; ; \; (T \to T_c^-, p = p_c)$$

$$\beta = \frac{1}{2}$$

2. 临界指数 δ

$$H = m^{\delta} \; ; \; (T = T_c, H \to 0)$$

$$(p - p_c) \sim (\rho - \rho_c)^{\delta} \; ; \; (T = T_c, p \to p_c)$$

$$\delta = 3$$

3. 临界指数 γ

$$\chi^0 = \left(\frac{\partial m}{\partial H}\right)_T \Big|_{H\to 0} \sim |T - T_c|^{-\gamma} \; ; \; (T \to T_c, H \to 0)$$

$$\kappa_T \sim |T - T_c|^{-\gamma} \; ; \; (T \to T_c^-, p = p_c)$$

$$\gamma = 1$$

4. 临界指数 α

$$C_H^0 \sim |T - T_c|^{-\alpha} \; ; \; (T \to T_c, H = 0)$$

$$C_V \sim |T - T_c|^{-\alpha} \; ; \; (T \to T_c, p = p_c)$$

$$\alpha = 0$$